Digital Twin Virtual Physical Model

Development Plan for Cock / Poultry Farm Manufacturing Operations

Planning draft | Prepared for software development estimation and product buildout | June 2026

1. Executive Summary

This document defines a practical development plan for a digital twin system that represents a poultry/cock farm as a synchronized virtual physical model. The system will connect real farm equipment, sensors, production records, and operational workflows into a 3D dashboard for monitoring, simulation, optimization, and reporting.

The recommended first release should not try to control every actuator automatically. It should start with reliable data collection, asset modeling, alarms, and operator visibility. After live telemetry is stable, the platform can add what-if simulation, predictive maintenance, animal health analytics, and guarded human-approved control actions.

- **Primary business goal:** reduce production loss by detecting environment, equipment, feed, water, and health problems earlier.
- **Technical goal:** maintain a live twin of houses, flocks, sensors, equipment, and production KPIs with traceable historical data.
- **Operational goal:** give farm managers one screen for real-time status, alarms, trends, and recommended actions.
- **Product goal:** build a scalable platform that can support one house first, then multiple houses, farms, hatchery/feed-mill modules, and external integrations.

2. Project Assumptions and Boundaries

Area	Assumption / Boundary
Terminology	"Cock farm manufacturing" is interpreted as a poultry/rooster farm production operation with barns/houses, flock batches, feeding, watering, ventilation, health monitoring, and production workflows.
Safety	Phase 1 and Phase 2 should be read-only monitoring and recommendation-driven. Any automatic actuator control should be added only after safety rules, operator approval, and field validation.
Hardware dependency	Final connector design depends on actual PLC/SCADA model, sensor models, protocols, farm network, and equipment vendor documentation.
Regulation and welfare	The development team must confirm local animal welfare, food safety, worker safety, data privacy, and electrical/industrial automation requirements before go-live.
3D model accuracy	The 3D model does not need CAD-level precision for MVP. It must be operationally accurate: houses, zones, equipment locations, sensor locations, and alarm overlays should be clear.

3. Objectives and Success KPIs

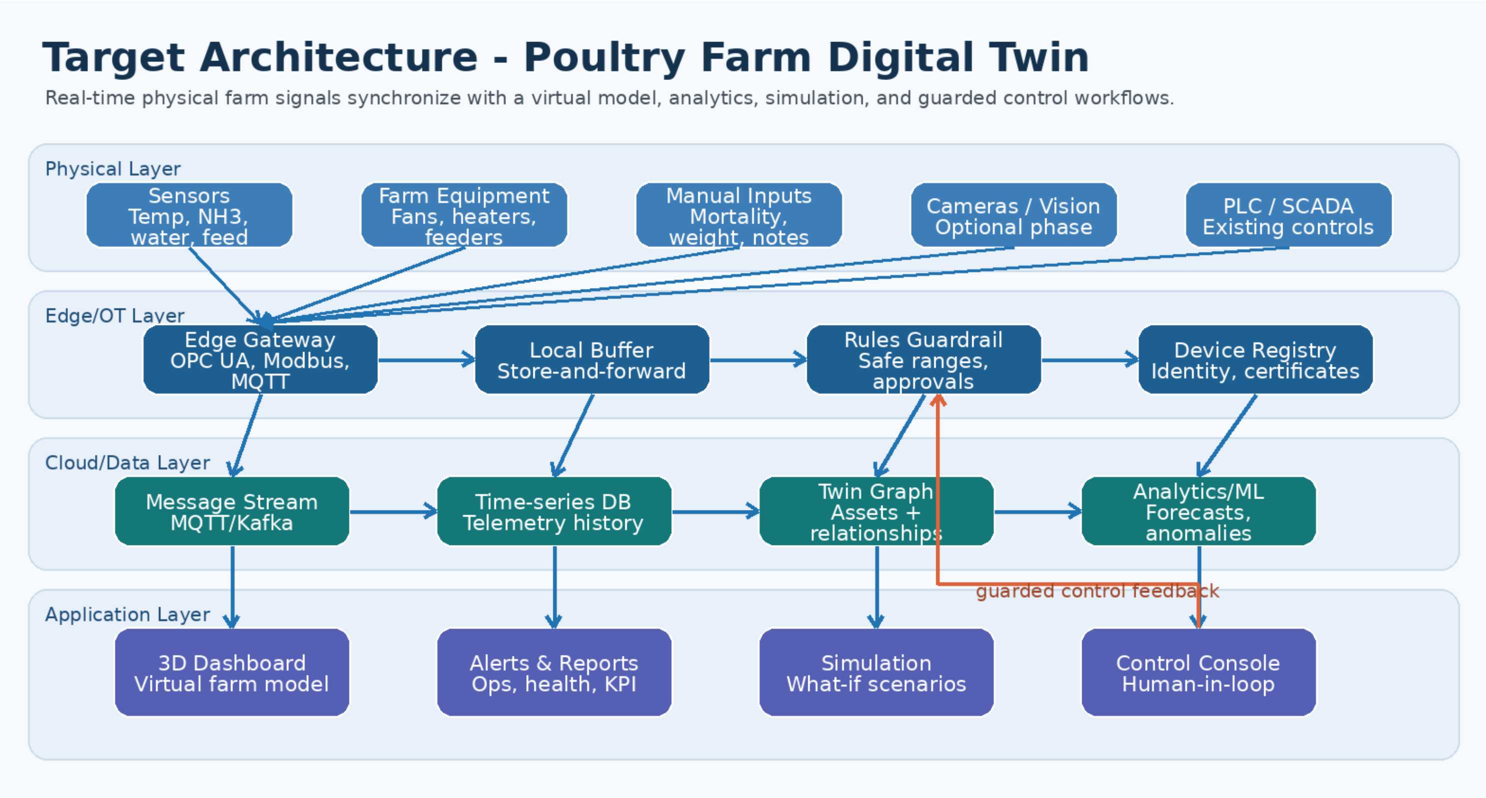
Objective	KPI / Measurement	Target for First Production Release
Real-time visibility	Telemetry freshness, dashboard latency	95% of critical signals visible within 5-10 seconds of edge receipt
Environment stability	Temperature, humidity, ammonia, CO2, air velocity excursions	Detect and alert abnormal conditions within 1 minute
Equipment reliability	Fan/feeder/water-line motor faults, runtime, current draw, downtime	Track equipment state and downtime with event history
Production performance	Feed conversion ratio, weight gain, mortality rate, water/feed usage per bird	Daily KPI report by house, flock batch, and farm
Predictive insight	Forecasted risk of high temperature, water drop, feeder failure, abnormal mortality	Pilot at least 3 high-value prediction models after baseline data is collected
Operator adoption	Daily active usage, alarm acknowledgement, report export usage	Farm manager uses dashboard every day for house review

4. Core System Scope

- **Asset hierarchy:** Farm > House > Zone/Pen > Equipment > Sensor > Telemetry tag.
- **3D virtual model:** Interactive 3D farm/house view with overlays for environment, equipment status, animal health, and alarms.
- **Real-time monitoring:** Temperature, humidity, ammonia, CO2, air velocity, water pressure/flow, feed level, light intensity, equipment state, and power usage.
- **Production management:** Flock batch, breed, age, stocking density, feed consumption, mortality, average weight, vaccination/task notes, and daily performance.
- **Alarms and recommendations:** Threshold alerts, trend-based alerts, operator acknowledgement, recommended corrective actions, escalation rules.
- **Simulation:** What-if scenarios such as ventilation increase, feed-line failure, heat-wave response, stocking density change, or energy-saving control strategy.
- **Analytics:** Historical charts, KPI cards, health distribution, equipment runtime, energy monitoring, reports, and prediction models.

- **Integrations:** PLC/SCADA, IoT gateways, sensor APIs, manual mobile forms, ERP/accounting feed purchase data, and optional camera/vision services.

5. Recommended Architecture



Architecture principles:

- **Separate IT and OT:** Never allow cloud software to directly override PLC safety logic. Cloud can recommend or request actions; critical control remains guarded at the edge/PLC layer.
- **Store raw and modeled data:** Keep raw telemetry for audit/debugging and also maintain a normalized twin model for the application.
- **Edge-first resilience:** The farm must continue operating during internet outages. The edge gateway buffers data and syncs when connectivity returns.
- **Event-driven design:** Telemetry events, alarms, predictions, and user actions should be published through a message bus to keep modules decoupled.
- **Model-first design:** Every screen, alert, simulation, and report should reference a real asset or flock batch in the twin graph.

6. Main Modules

Module	Responsibility	Suggested Implementation
Twin Model Service	Stores asset hierarchy, equipment metadata, sensor mapping, relationships, spatial coordinates, and status.	PostgreSQL + PostGIS or Neo4j; optional Azure Digital Twins/AWS IoT TwinMaker
Telemetry Ingestion	Receives sensor/PLC data, validates tags, normalizes units, writes time-series records, publishes events.	MQTT, OPC UA, Modbus TCP, Kafka/Redpanda, TimescaleDB/InfluxDB
Edge Gateway	Runs inside farm network, connects to PLC/SCADA/sensors, buffers offline data, applies safety guardrails.	Industrial PC, Docker, Node-RED/Python/FastAPI, local SQLite/Timescale buffer

3D Dashboard	Shows virtual farm model, house cutaway, equipment overlays, alarms, trends, and simulation results.	React/Next.js, Three.js or Babylon.js, WebSocket/SSE
Production Records	Captures flock batch, age, weight, mortality, feed, water, medicine/vaccination notes, and daily reports.	Web forms, mobile-friendly PWA, PostgreSQL
Alarm Engine	Threshold, trend, correlation, equipment fault, and predictive alarms with acknowledgement workflow.	Rules engine + job queue + notification service
Simulation Engine	Runs what-if scenarios using farm geometry, sensor history, equipment capacity, and simple physics/ML models.	Python services, SimPy/custom model, optional external simulator
Analytics/ML	Detects anomalies, forecasts risk, predicts equipment maintenance, and provides recommendations.	Python, scikit-learn/PyTorch, MLflow, feature store
Admin/Security	RBAC, tenant/farm permissions, audit logs, device identity, API keys/certificates, backups.	OAuth/OIDC, JWT/session, TLS certificates, audit tables

7. Data Model - First Version

Entity	Key Fields	Purpose
Farm	id, name, location, timezone, owner, status	Top-level tenant/operation unit.
House	id, farm_id, name, dimensions, capacity, current_flock_id	Physical poultry house/barn.
Zone / Pen	id, house_id, coordinates, capacity, equipment_group	Spatial subdivision used for heat maps and alarms.
FlockBatch	id, house_id, breed, start_date, age_days, initial_count, current_count	Production batch; links all daily records.
Equipment	id, type, model, vendor, zone_id, protocol, plc_tag_prefix	Fans, feeders, drinkers, heaters, pumps, belts, meters.
Sensor	id, equipment_id/zone_id, tag, unit, calibration, min/max	Physical sensor or PLC tag mapped into the twin.
TelemetryPoint	sensor_id, timestamp, value, quality, source	Time-series measurement. Use partitioning/retention policy.
Alarm	asset_id, severity, rule_id, status, opened_at, acknowledged_by, closed_at	Event lifecycle and operator audit trail.
DailyProductionRecord	flock_id, date, feed_used, water_used, mortality, avg_weight, notes	Manual and automatic daily production KPIs.
SimulationScenario	name, parameters, baseline_range, result, created_by	Saved what-if runs and expected outcomes.

8. Sensor and Integration Plan

Signal / Device	Purpose	Typical Source	Priority
Temperature and humidity	Climate monitoring and heat/cold stress detection	Environmental sensors / PLC	MVP
Ammonia (NH3) and CO2	Air quality and ventilation risk	Gas sensors	MVP if hardware available
Water pressure and flow	Detect water-line blockage, low intake, leaks, pump issues	Flow meter / pressure sensor	MVP

Feed silo level / load cell	Feed inventory and feed conversion calculation	Silo sensor / manual entry	MVP or Phase 2
Feeder motor current/status	Detect feeder jam/fault and runtime	PLC tag / current sensor	MVP
Fan/heater status and runtime	Ventilation/heating performance and maintenance	PLC/SCADA	MVP
Light intensity	Lighting program verification	Lux sensor / controller API	Phase 2
Bird weight scale	Growth curve and abnormal growth detection	Automatic scale / manual sample	Phase 2
Camera/vision	Density, behavior, dead bird detection, equipment view	IP camera + AI model	Phase 3
Weather	External heat/rain risk and ventilation context	Weather station/API	Phase 2

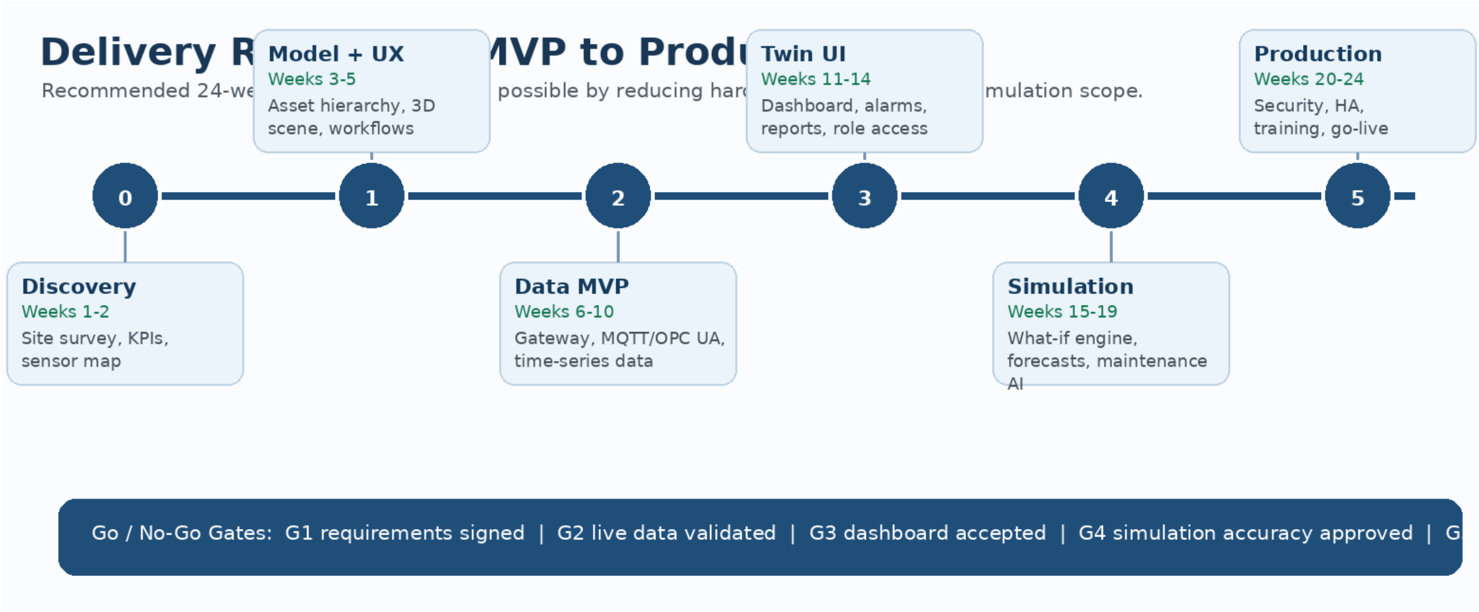
Integration protocols to support:

- MQTT for lightweight IoT telemetry and device-to-cloud messages.
- OPC UA for industrial interoperability with PLC/SCADA systems when supported.
- Modbus TCP/RTU for common industrial sensors and controllers.
- REST/Webhook imports for manual production systems, ERP, or equipment vendor APIs.
- CSV/Excel import for historical baseline data during pilot setup.

9. User Experience and Screens

Screen	Core Content	Main User Action
Overview dashboard	All farms/houses, live status, alarms, KPI cards, environment summary	Identify which house needs attention
3D house twin	Cutaway model, equipment locations, sensor overlays, alarm pins, status labels	Click asset/zone to inspect details
House details	Temperature map, air quality, feed/water trends, equipment status, flock summary	Review house condition and acknowledge alarms
Production daily entry	Mortality, weight, feed, water, notes, health observations	Submit or correct daily records
Equipment details	Runtime, failure history, maintenance schedule, motor current, parts notes	Create maintenance task
Simulation	Scenario selection, parameters, expected effects, comparison with baseline	Run what-if and save recommendation
Reports	Daily/weekly PDF or Excel export by farm, house, flock, and date range	Share report with owner/manager
Admin	Users, roles, farms, houses, sensors, equipment, alarm rules, thresholds	Configure the model and permissions

10. Development Roadmap



Phase	Duration	Deliverables	Acceptance Criteria
Phase 0 - Discovery	1-2 weeks	Site survey, farm layout, sensor/protocol inventory, KPI list, risk log, MVP scope.	Requirements signed off; available data sources and missing hardware are clear.
Phase 1 - Twin model and UX prototype	2-3 weeks	Asset hierarchy, database schema, wireframes, 3D scene prototype, API contract.	Stakeholders can navigate a sample house and understand all core screens.
Phase 2 - Data ingestion MVP	4-5 weeks	Edge gateway, MQTT/OPC UA connectors, telemetry normalization, time-series storage, live status API.	At least 20-50 live tags flow from pilot house with quality flags and reconnect behavior.
Phase 3 - Dashboard and alarm MVP	3-4 weeks	3D dashboard, KPI cards, trend charts, alarm rules, acknowledgement workflow, daily production entry.	Farm manager can use the dashboard for daily operation review.
Phase 4 - Simulation and analytics	4-5 weeks	What-if scenario engine, baseline comparison, anomaly detection, equipment prediction pilot, recommendation cards.	At least 3 scenarios and 3 predictive insights are tested against real historical/live data.
Phase 5 - Production hardening	4-5 weeks	RBAC, audit logs, HA deployment, backups, observability, security review, training, go-live runbook.	Production readiness checklist passes and pilot farm goes live.

11. Team Plan

Role	Responsibility	Suggested Allocation
Product / Domain Lead	Define farm workflows, KPIs, alarm logic, production reporting, stakeholder acceptance.	0.5-1 FTE through discovery and acceptance
Solution Architect	Architecture, IT/OT boundary, cloud/edge design, data model, security decisions.	0.5 FTE throughout
Backend Engineer	APIs, telemetry pipeline, twin service, production records, alarm engine.	1-2 FTE
Frontend / 3D Engineer	React dashboard, 3D model integration, charts, real-time overlays, admin UI.	1-2 FTE

IoT / Edge Engineer	Gateway setup, PLC/SCADA integration, protocols, device identity, offline buffering.	1 FTE during Phases 0-3
Data / ML Engineer	Forecasting, anomaly detection, simulation models, feature pipelines.	0.5-1 FTE from Phase 4
QA Engineer	Test plans, simulator tests, integration tests, performance tests, regression testing.	0.5-1 FTE from Phase 2
DevOps / Security	CI/CD, infrastructure, monitoring, backups, secrets, network security.	0.5 FTE from Phase 2

12. Technology Stack Options

Layer	Recommended Custom Stack	Managed Platform Option	Notes
Frontend	Next.js/React, TypeScript, Three.js/Babylon.js, ECharts/Recharts	Same frontend; embed managed twin viewer if selected	Use WebSocket/SSE for live cards and overlays.
Backend APIs	FastAPI or NestJS, PostgreSQL, Redis, REST/WebSocket	AWS Lambda/ECS or Azure App Service/Container Apps	FastAPI is strong for Python analytics; NestJS is strong for TypeScript teams.
Time-series data	TimescaleDB or InfluxDB	AWS Timestream, Azure Data Explorer	TimescaleDB simplifies SQL analytics with PostgreSQL.
Message layer	MQTT broker + Kafka/Redpanda for stream processing	AWS IoT Core / Azure IoT Hub	MQTT is useful at edge; Kafka-like stream is useful inside backend.
Twin graph	PostgreSQL graph-style tables or Neo4j	Azure Digital Twins or AWS IoT TwinMaker	Managed platforms reduce twin-model boilerplate but may increase cloud lock-in.
3D model	GLB/GLTF assets, Three.js/Babylon.js scene graph	AWS IoT TwinMaker scene / Azure 3D Scenes Studio	Start simple: house shell, equipment icons, zones, labels.
ML/simulation	Python, Pandas, scikit-learn, PyTorch, MLflow, SimPy/custom models	SageMaker / Azure ML	Begin with interpretable rules and trend models before complex AI.
Deployment	Docker, Kubernetes or Docker Compose for pilot, Terraform, GitHub Actions	AWS/Azure managed containers	Pilot can run simpler; production needs backup, observability, and HA.

13. MVP Feature Set

MVP Feature	Included	Not Included Yet
Farm and house setup	Create farm, houses, zones, equipment, sensors, and flock batch.	Complex multi-tenant billing or franchise management.
Live telemetry	Ingest temperature, humidity, water, feed, fan/feeder state, and equipment status from pilot house.	Full integration for every vendor device on day one.
3D virtual model	Operationally accurate house cutaway with equipment/sensor overlays.	Photorealistic CAD/reality-capture model.
Alarms	Threshold alarms, acknowledgement, history, simple recommendation text.	Fully autonomous closed-loop control.
Production records	Daily mortality, feed, water, average weight, notes, report export.	Advanced veterinary diagnosis or computer vision.
Analytics	Trend charts, KPI cards, daily/weekly reports.	Advanced predictive AI beyond pilot rules.
Simulation	Basic what-if for ventilation and environment response.	High-fidelity CFD or full animal behavior simulation.

14. Production Hardening Checklist

- **Reliability:** edge buffering, automatic reconnect, duplicate detection, idempotent ingestion, backfill jobs, queue monitoring.
- **Security:** TLS everywhere, device certificates or strong API keys, VPN/private networking for OT access, RBAC, audit logs, secret rotation.
- **Observability:** metrics, logs, traces, data freshness dashboard, alarm delivery monitoring, failed ingestion alerts.
- **Data governance:** retention policy, backup and restore test, unit normalization, calibration history, source quality flags.
- **Performance:** load test for live telemetry rate, dashboard concurrent users, report generation, and historical chart queries.
- **Safety:** manual approval for control actions, safe setpoint bounds, emergency override, local PLC safety logic, operator training.
- **Operations:** runbooks, incident response, maintenance schedule, training materials, go-live checklist, rollback plan.

15. Key Risks and Mitigations

Risk	Impact	Mitigation
Sensor data is incomplete or unreliable	Dashboard trust becomes low and analytics fail.	Start with data audit; show data quality flags; calibrate sensors; allow manual fallback entry.
PLC/SCADA integration is blocked by vendor or protocol limits	Live equipment data cannot be captured.	Use read-only gateway first; request vendor docs early; support MQTT, OPC UA, Modbus, REST, and CSV alternatives.
3D model scope grows too much	Timeline delays and visual work distracts from operational value.	Define MVP visual accuracy level; use simple GLB assets; add photorealism later.
Automatic control creates safety risk	Animal welfare, equipment, or worker safety issues.	No direct auto-control in MVP; use recommendations and human approval; keep PLC safety logic independent.
Prediction models lack enough data	AI recommendations are weak or misleading.	Use rules and baseline statistics first; collect 2-3 flock cycles before advanced ML.
Farm network is unstable	Data gaps and user frustration.	Edge store-and-forward, local dashboard fallback, network survey, offline indicators.
Operators do not adopt the dashboard	Business value is not realized.	Design for daily farm routine; keep data entry minimal; add clear alarm/recommendation workflow.

16. Estimation Guidance

Build Option	Timeline	Estimated Team	Best For
Clickable concept / investor demo	2-4 weeks	1 frontend/3D + 1 backend part-time	Showing vision, mock data, pitch/demo.
Pilot MVP for one house	10-14 weeks	Frontend, backend, IoT/edge, product/domain, QA part-time	Real telemetry, basic dashboard, alarms, daily records.
Production v1 for one farm	20-24 weeks	Full small team: 4-7 people including IoT and QA	Operational use, security, reporting, training, stable deployment.
Multi-farm commercial platform	6-12 months+	Product team plus engineering team	SaaS product, tenant management, advanced analytics, mobile app, integrations.

Cost drivers are hardware integration complexity, number of sensor/PLC vendors, required 3D fidelity, simulation accuracy, offline requirements, and whether managed cloud digital twin services are used.

17. Immediate Next Steps

1. **Confirm pilot site:** choose one farm and one house as the first implementation target.
2. **Collect layout and equipment data:** house dimensions, equipment list, sensor list, PLC/SCADA details, network diagram, existing reports.
3. **Define MVP KPIs:** select the top 10-15 operational signals and 5-8 business KPIs for the first dashboard.
4. **Choose stack path:** custom stack, Azure Digital Twins, AWS IoT TwinMaker, or hybrid.
5. **Build prototype:** create asset hierarchy, sample 3D house, mock data dashboard, and daily production form.
6. **Run telemetry pilot:** connect edge gateway to a small set of real signals, validate data quality, then expand.
7. **Finalize production plan:** turn pilot learnings into hardware list, implementation schedule, QA plan, and go-live checklist.

18. Reference Basis

These sources were used to align the plan with common industrial digital twin concepts and platform options:

Ref	Source	Planning Use	URL
[1]	AWS IoT TwinMaker	AWS describes IoT TwinMaker as a service for creating digital twins of real-world systems to monitor and optimize industrial operations.	https://aws.amazon.com/iot-twinmaker/
[2]	AWS Industrial Digital Twin Guidance	AWS guidance highlights immersive 3D views of systems and operations for efficiency, production, and performance optimization.	https://docs.aws.amazon.com/solutions/industrial-digital-twin-on-aws/
[3]	Azure Digital Twins	Microsoft describes Azure Digital Twins as a PaaS for twin graphs based on digital models of environments such as factories and farms.	https://learn.microsoft.com/en-us/azure/digital-twins/overview
[4]	OPC UA / OPC Foundation	OPC UA is widely used for secure and reliable industrial data exchange and is standardized as IEC 62541.	https://opcfoundation.org/opcua-en.pdf

Note: This plan is a software/product planning document. Hardware selection, electrical installation, animal welfare controls, and safety validation require qualified local specialists and equipment vendor input.